Dataset View - Custom

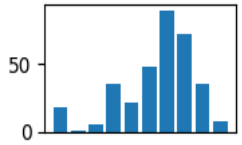
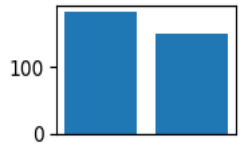
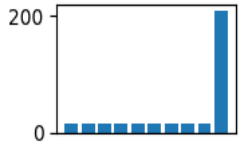
Dataset Head

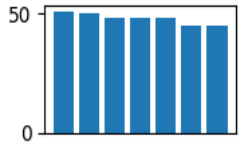
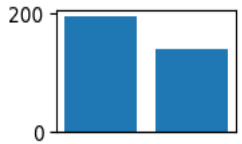
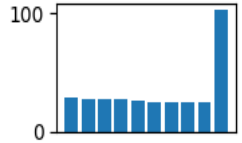
age	gender	pid	diagnosis	present	gxd	age_bin
65.0	F	patient 1	Acute MI	NO	F_Acute MI	60-70
65.0	F	patient 1	Transient ischemic attack	NO	F_Transient ischemic attack	60-70
65.0	F	patient 1	left ventricular hypertrophy	NO	F_left ventricular hypertrophy	60-70
65.0	F	patient 1	Earlier MI	NO	F_Earlier MI	60-70
65.0	F	patient 1	Sinus node dysfunction	NO	F_Sinus node dysfunction	60-70
59.0	M	patient 2	Coronary artery disease	NO	M_Coronary artery disease	50-60
59.0	M	patient 2	arterial hypertension	NO	M_arterial hypertension	50-60
59.0	M	patient 2	Transient ischemic attack	NO	M_Transient ischemic attack	50-60
59.0	M	patient 2	left ventricular hypertrophy	NO	M_left ventricular hypertrophy	50-60
59.0	M	patient 2	Earlier MI	NO	M_Earlier MI	50-60
59.0	M	patient 2	Sinus node dysfunction	NO	M_Sinus node dysfunction	50-60
80.0	M	patient 3	Coronary artery disease	NO	M_Coronary artery disease	70-80
80.0	M	patient 3	arterial hypertension	NO	M_arterial hypertension	70-80
80.0	M	patient 3	Acute MI	NO	M_Acute MI	70-80
80.0	M	patient 3	left ventricular hypertrophy	NO	M_left ventricular hypertrophy	70-80
80.0	M	patient 3	Earlier MI	NO	M_Earlier MI	70-80
80.0	M	patient 3	Sinus node dysfunction	NO	M_Sinus node dysfunction	70-80
51.0	F	patient 4	Coronary artery disease	NO	F_Coronary artery disease	50-60
51.0	F	patient 4	arterial hypertension	NO	F_arterial hypertension	50-60
51.0	F	patient 4	Acute MI	NO	F_Acute MI	50-60
51.0	F	patient 4	Transient ischemic attack	NO	F_Transient ischemic attack	50-60
51.0	F	patient 4	left ventricular hypertrophy	NO	F_left ventricular hypertrophy	50-60
51.0	F	patient 4	Earlier MI	NO	F_Earlier MI	50-60
51.0	F	patient 4	Sinus node dysfunction	NO	F_Sinus node dysfunction	50-60
68.0	M	patient 5	Coronary artery disease	NO	M_Coronary artery disease	60-70
68.0	M	patient 5	arterial hypertension	NO	M_arterial hypertension	60-70
68.0	M	patient 5	Acute MI	NO	M_Acute MI	60-70

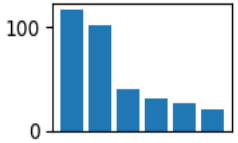
68.0	M	patient 5	Transient ischemic attack	NO	M_Transient ischemic attack	60-70
68.0	M	patient 5	left ventricular hypertrophy	NO	M_left ventricular hypertrophy	60-70
68.0	M	patient 5	Earlier MI	NO	M_Earlier MI	60-70

Features View

Stats

feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
age	0	18.0	82.0499 7905941 605	56.2641 6839909 784	13.4939 2751295 0515	50.4618 1516322 757	59.0	65.9537 1601122 292	float			False	False
gender	0								binary	F, M		True	False
pid	0								categori cal	p0, p1, p2, p3, p4, p5, p6, p7, p8, p9, patient 1, patient 10, patient 11, patient 12, patient 13, patient 14, patient 15, patient 16, patient 17, pat...		False	False

feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
diagnosis	0								categorical	Acute MI, Coronary artery disease, Earlier MI, Sinus node dysfunction, Transient ischemic attack, arterial hypertension, left ventricular hypertrophy		True	False
present	0								binary	NO, YES		False	True
gxd	0								categorical	F_Acute MI, F_Coronary artery disease, F_Earlier MI, F_Sinus node dysfunction, F_Transient ischemic attack, F_arterial hypertension, F_left ventricula...		True	False

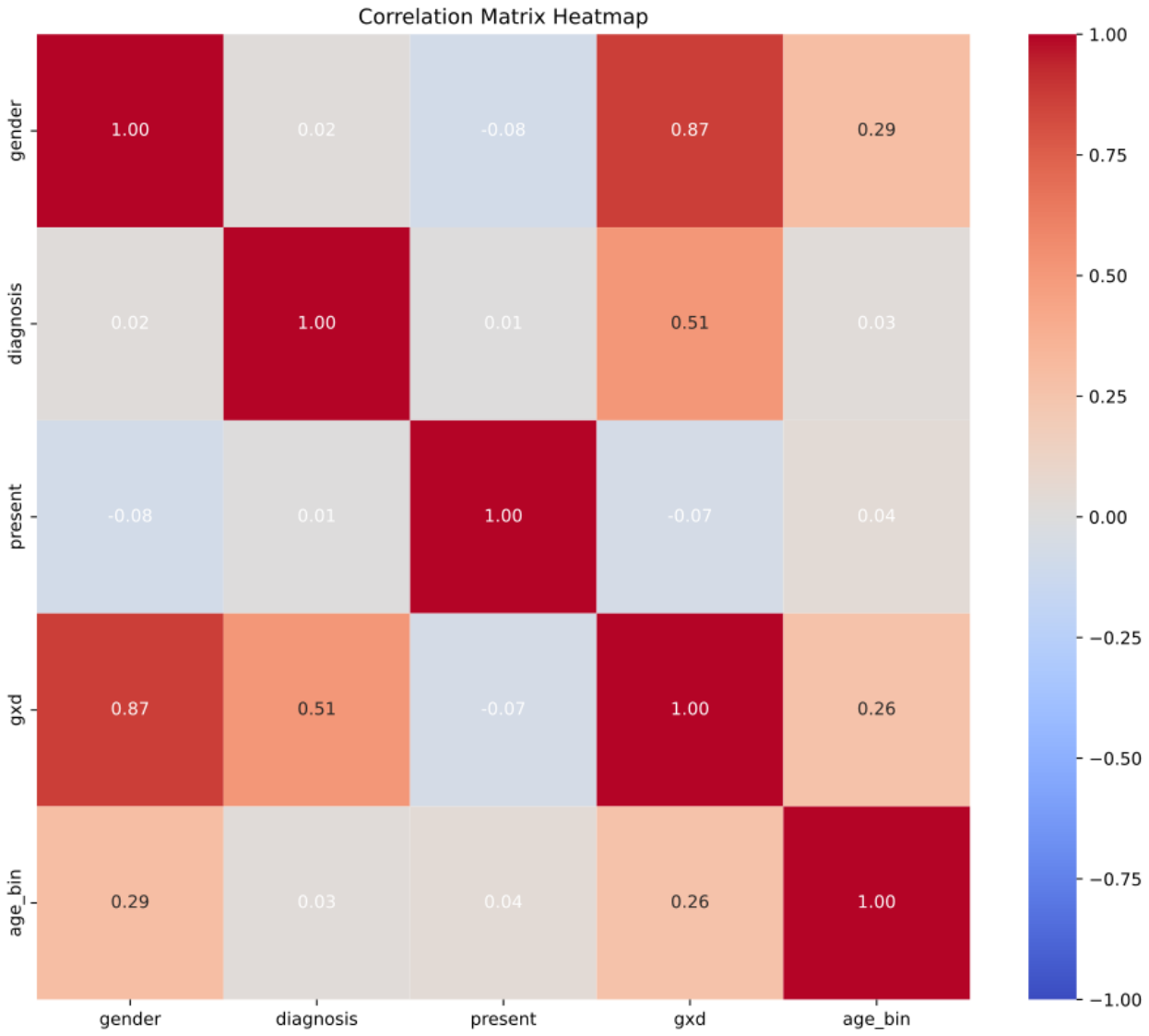
feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
age_bin	0								categorical	0-20, 30-40, 40-50, 50-60, 60-70, 70-80		True	False

Features

Field Name	drop	sensitive	target
age	✔	✘	✘
gender	✘	✔	✘
pid	✔	✘	✘
diagnosis	✘	✔	✘
present	✘	✘	✔
gxd	✘	✔	✘
age_bin	✘	✔	✘

Proxies

Correlation Matrix



Suggested Proxies

Gender

	diagnosis	gxd	age_bin
correlation	1.59%	86.93%	29.05%
suggested_proxy	✗	✓	✗

Diagnosis

	gender	gxd	age_bin
correlation	1.59%	50.80%	2.82%
suggested_proxy	✗	✗	✗

Gxd

	gender	diagnosis	age_bin
correlation	86.93%	50.80%	26.42%
suggested_proxy	✓	✗	✗

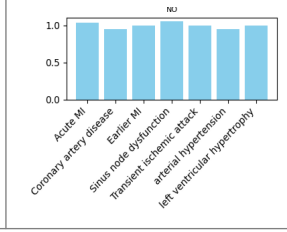
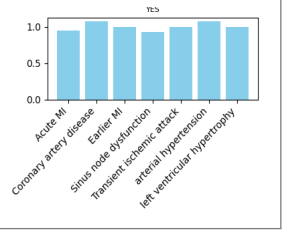
Age_bin

	gender	diagnosis	gxd
correlation	29.05%	2.82%	26.42%
suggested_proxy	✗	✗	✗

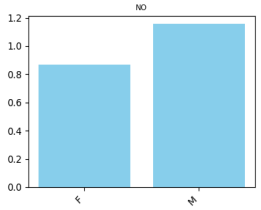
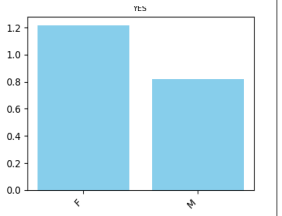
Detection

Selected Metrics

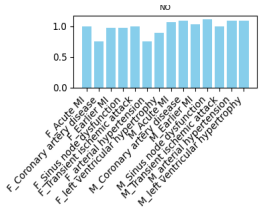
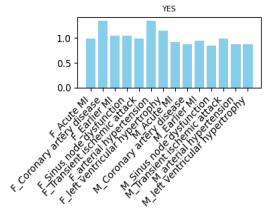
DisparateImpact

	NO	YES
diagnosis		

DisparateImpact

	NO	YES
gender		

DisparateImpact

	NO	YES
gxd		

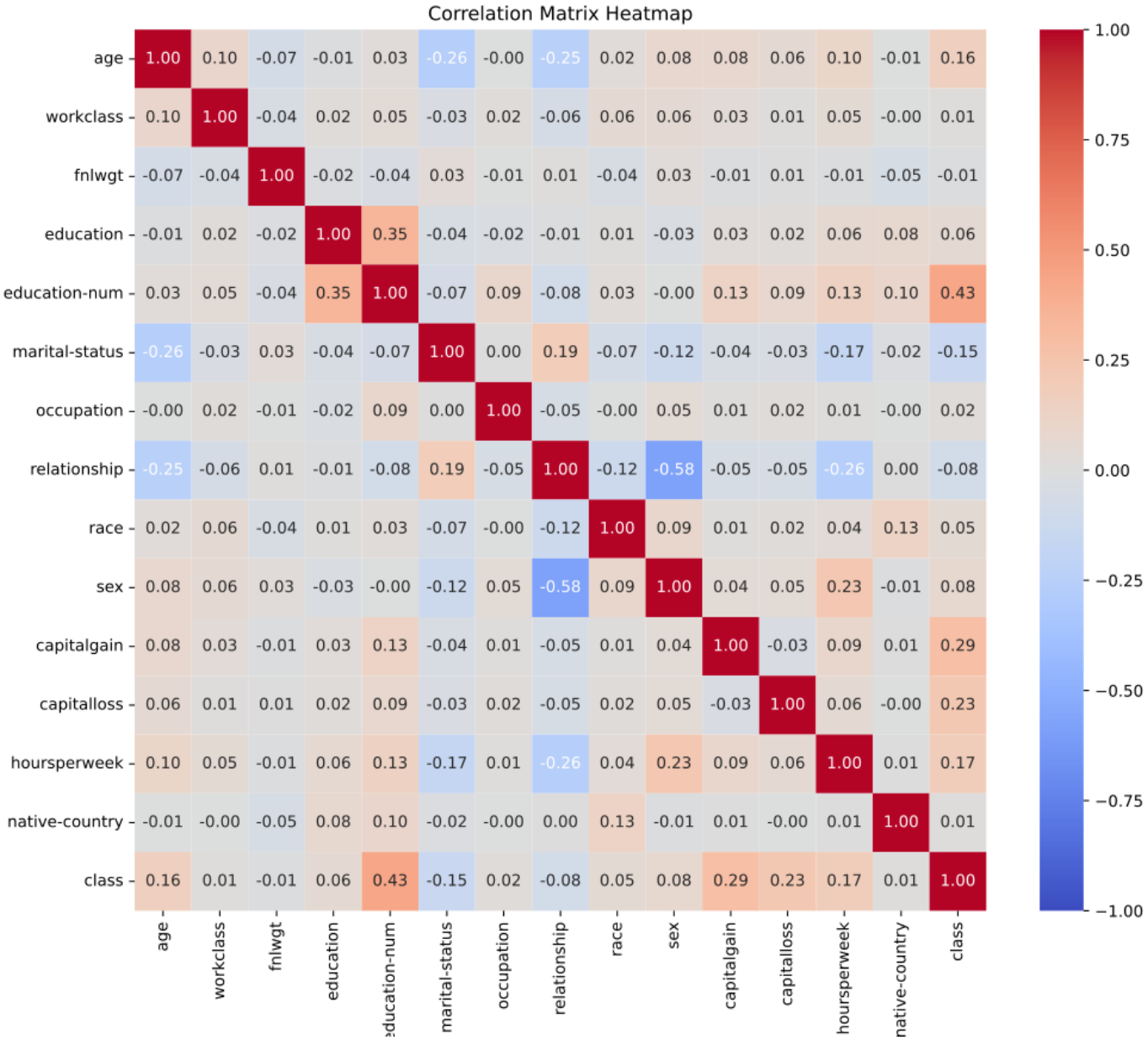
Model Mitigation

Inprocessing

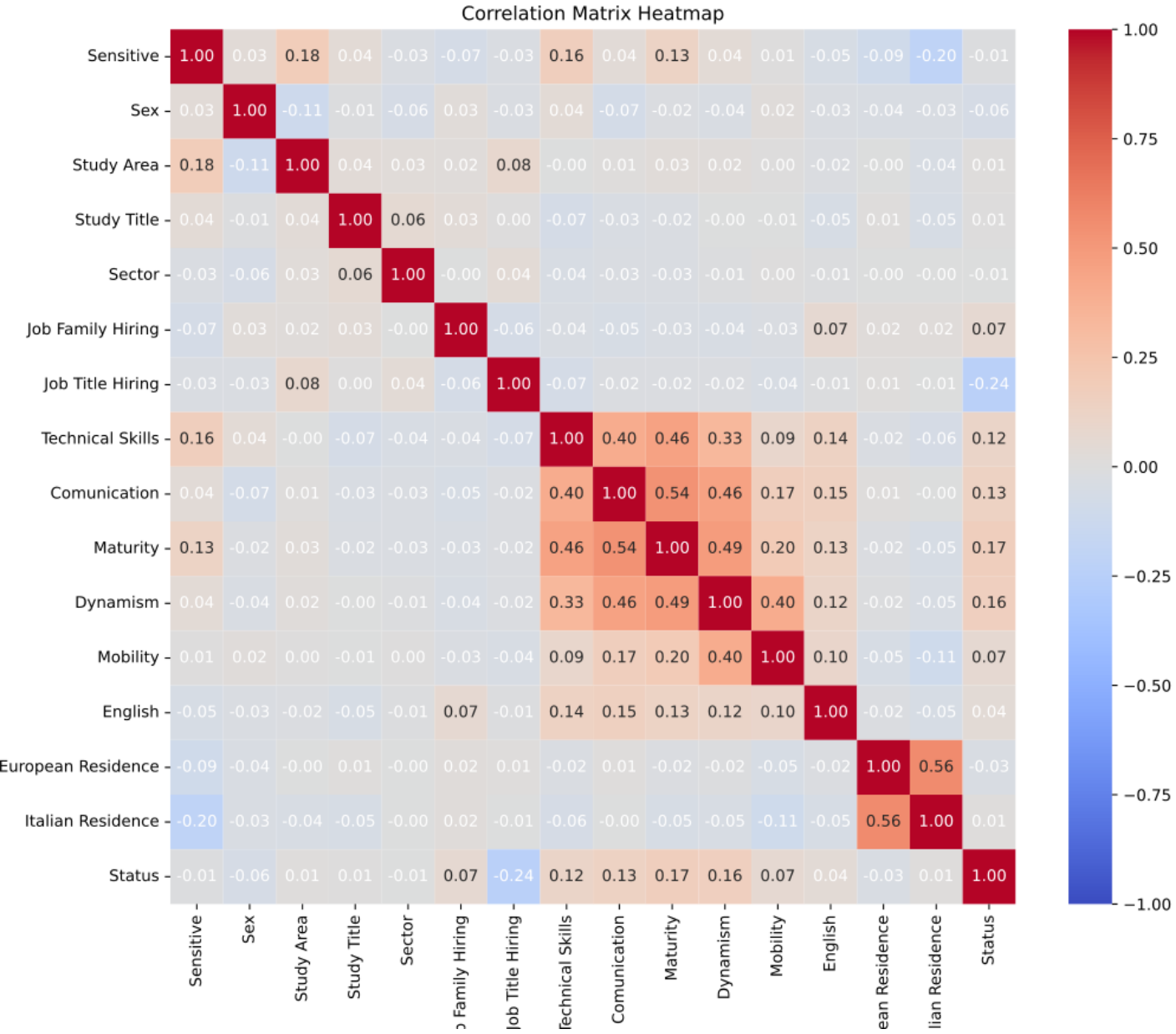
$algorithm	lambda
FaUCI	50.00%

Model Mitigation Results

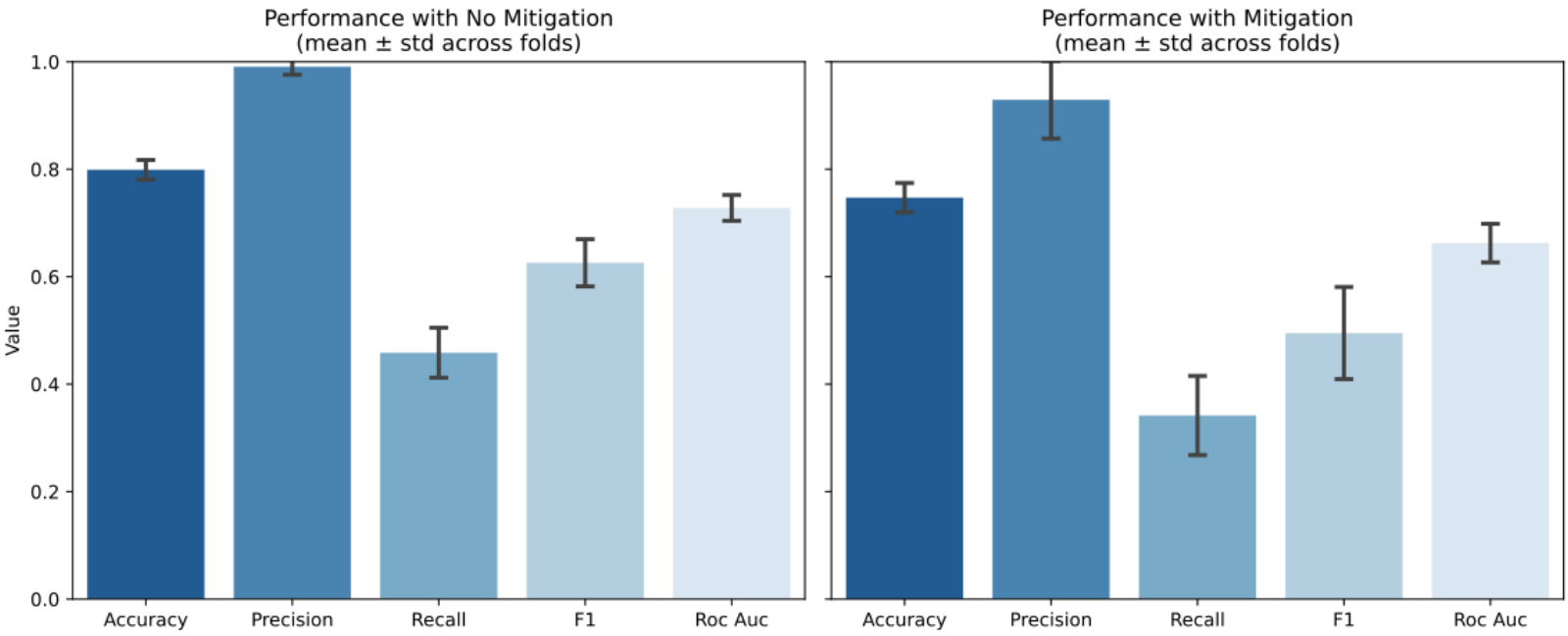
Correlation Matrix



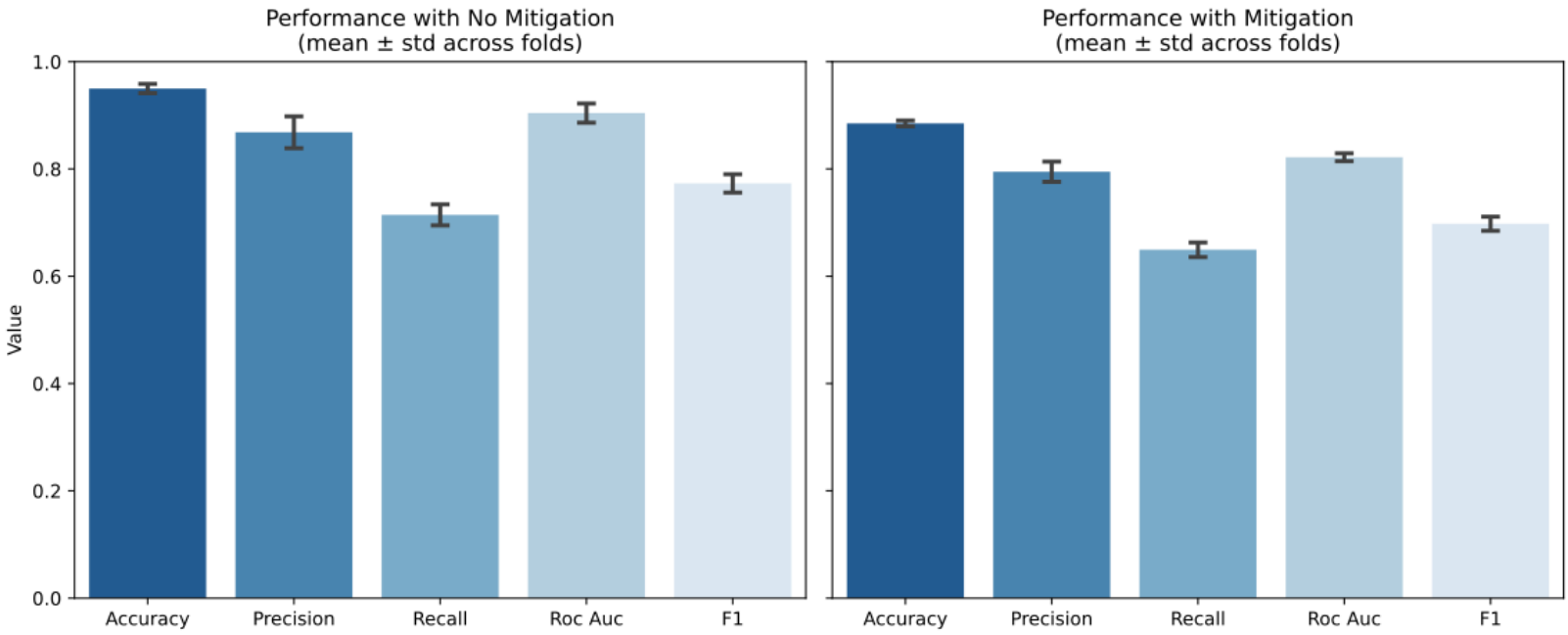
Correlation Matrix



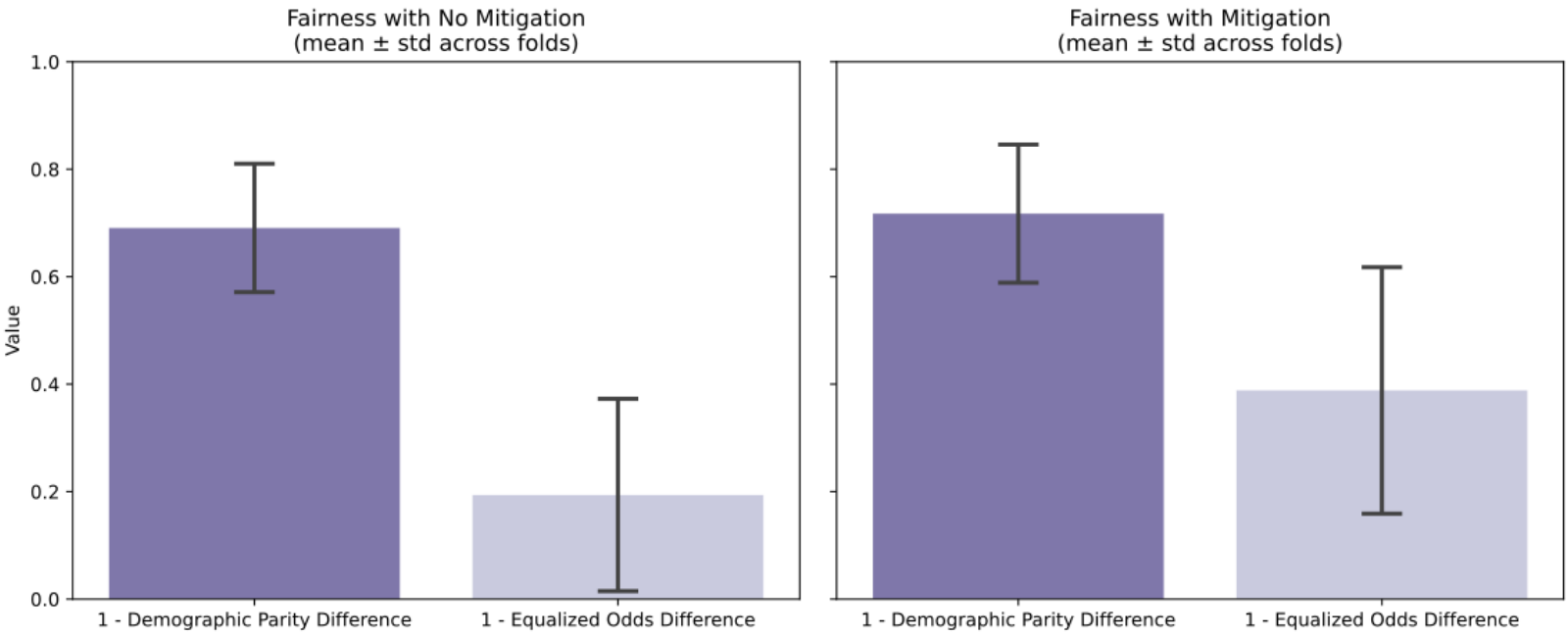
Performance Plot



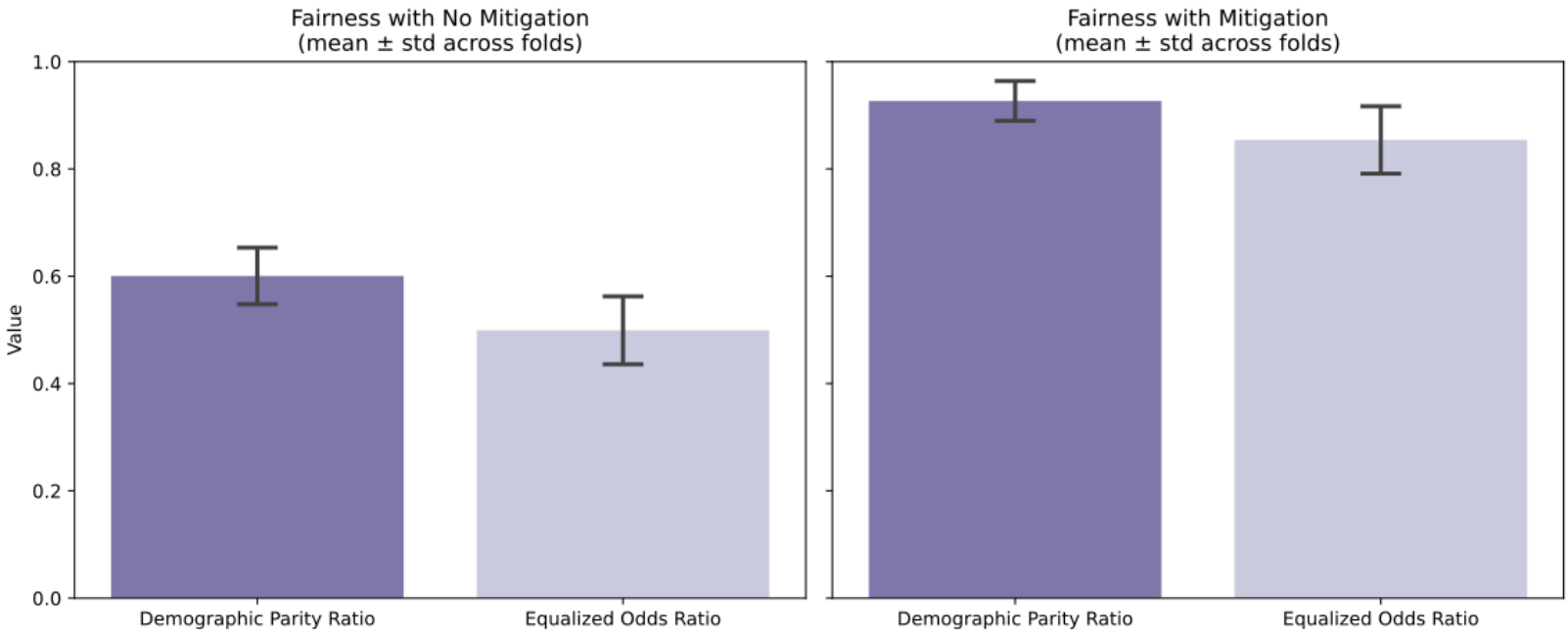
Performance Plot



Fairness Plot



Fairness Plot



Predictions Head

age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex	capitalgain	capitalloss	hoursperweek	native-country	class
25	Private	226802	11th	7	Never-married	Machine-op-inspct	Own-child	Black	Male	0	0	40	United-States	<=50K
38	Private	89814	HS-grad	9	Married-civ-spouse	Farming-fishing	Husband	White	Male	0	0	50	United-States	<=50K
28	Local-gov	336951	Assoc-acdm	12	Married-civ-spouse	Protective-serv	Husband	White	Male	0	0	40	United-States	<=50K
44	Private	160323	Some-college	10	Married-civ-spouse	Machine-op-inspct	Husband	Black	Male	7688	0	40	United-States	>50K
34	Private	198693	10th	6	Never-married	Other-service	Not-in-family	White	Male	0	0	30	United-States	<=50K
63	Self-emp-not-inc	104626	Prof-school	15	Married-civ-spouse	Prof-specialty	Husband	White	Male	3103	0	32	United-States	>50K
24	Private	369667	Some-college	10	Never-married	Other-service	Unmarried	White	Female	0	0	40	United-States	<=50K
55	Private	104996	7th-8th	4	Married-civ-spouse	Craft-repair	Husband	White	Male	0	0	10	United-States	<=50K
65	Private	184454	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	6418	0	40	United-States	<=50K
36	Federal-gov	212465	Bachelors	13	Married-civ-spouse	Adm-clerical	Husband	White	Male	0	0	40	United-States	<=50K
26	Private	82091	HS-grad	9	Never-married	Adm-clerical	Not-in-family	White	Female	0	0	39	United-States	<=50K
48	Private	279724	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	3103	0	48	United-States	<=50K
43	Private	346189	Masters	14	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0	50	United-States	>50K
20	State-gov	444554	Some-college	10	Never-married	Other-service	Own-child	White	Male	0	0	25	United-States	<=50K
43	Private	128354	HS-grad	9	Married-civ-spouse	Adm-clerical	Wife	White	Female	0	0	30	United-States	>50K
37	Private	60548	HS-grad	9	Widowed	Machine-op-inspct	Unmarried	White	Female	0	0	20	United-States	<=50K
34	Private	107914	Bachelors	13	Married-civ-spouse	Tech-support	Husband	White	Male	0	0	47	United-States	>50K

34	Private	238588	Some-college	10	Never-married	Other-service	Own-child	Black	Female	0	0	35	United-States	<=50K
25	Private	220931	Bachelors	13	Never-married	Prof-specialty	Not-in-family	White	Male	0	0	43	Peru	<=50K
25	Private	205947	Bachelors	13	Married-civ-spouse	Prof-specialty	Husband	White	Male	0	0	40	United-States	<=50K
45	Self-emp-not-inc	432824	HS-grad	9	Married-civ-spouse	Craft-repair	Husband	White	Male	7298	0	90	United-States	>50K
22	Private	236427	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0	20	United-States	<=50K
23	Private	134446	HS-grad	9	Separated	Machine-op-inspct	Unmarried	Black	Male	0	0	54	United-States	<=50K
54	Private	99516	HS-grad	9	Married-civ-spouse	Craft-repair	Husband	White	Male	0	0	35	United-States	<=50K
32	Self-emp-not-inc	109282	Some-college	10	Never-married	Prof-specialty	Not-in-family	White	Male	0	0	60	United-States	<=50K
46	State-gov	106444	Some-college	10	Married-civ-spouse	Exec-managerial	Husband	Black	Male	7688	0	38	United-States	>50K
56	Self-emp-not-inc	186651	11th	7	Widowed	Other-service	Unmarried	White	Female	0	0	50	United-States	<=50K
24	Self-emp-not-inc	188274	Bachelors	13	Never-married	Sales	Not-in-family	White	Male	0	0	50	United-States	<=50K
23	Local-gov	258120	Some-college	10	Married-civ-spouse	Protective-serv	Husband	White	Male	0	0	40	United-States	<=50K
26	Private	43311	HS-grad	9	Divorced	Exec-managerial	Unmarried	White	Female	0	0	40	United-States	<=50K

Predictions Head

Sensitive	Sex	Study Area	Study Title	Sector	Job Family Hiring	Job Title Hiring	Technical Skills	Comunication	Maturity	Dynamism	Mobility	English	European Residence	Italian Residence	Status
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Automotive	Engineering	Consultant	3	2	3	3	2	4	Yes	Yes	Positive
Not a protected category 40 - 45 years	Female	Engineering	Five-year degree	Aeronautics	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 36 - 40 years	Male	Engineering	Five-year degree	Consulting	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Positive
Not a protected category > 45 years	Male	Law	Five-year degree	Telecom	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Automotive	Not Specified	Not Specified	3	2	2	3	1	1	Yes	Yes	Negative
Not a protected category 36 - 40 years	Male	Engineering	Five-year degree	Telecom	Not Specified	Not Specified	2	2	3	3	3	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Female	Economic	Five-year degree	Others	Support	Support Advanced	2	2	2	2	2	2	Yes	Yes	Positive
Not a protected category 36 - 40 years	Male	Engineering	Five-year degree	Space	Not Specified	Not Specified	3	3	3	3	2	2	Yes	Yes	Negative

Not a protected category 36 - 40 years	Male	Scientific Field	Five-year degree	Life sciences	Engineering	Consultant	2	2	3	3	1	2	Yes	Yes	Positive
Not a protected category 40 - 45 years	Male	Engineering	Five-year degree	Unemployed	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 36 - 40 years	Male	Engineering	Five-year degree	Consulting	Engineering	Consultant	3	2	3	3	3	3	Yes	Yes	Positive
Not a protected category 36 - 40 years	Female	Engineering	Doctorate	Others	Engineering	Qualified Consultant	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Others	Not Specified	Not Specified	2	2	2	2	1	3	Yes	Yes	Negative
Not a protected category 40 - 45 years	Male	Engineering	Five-year degree	Others	Engineering	Qualified Consultant	3	2	2	2	2	2	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Consulting	Engineering	Consultant	2	2	2	2	2	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Telecom	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Others	Engineering	Consultant	3	3	3	2	2	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Consulting	Engineering	Consultant	3	3	3	2	2	2	Yes	Yes	Positive

Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Aeronautics	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Female	Engineering	Five-year degree	Unemployed	Engineering	Consultant	2	2	2	2	2	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Others	Engineering	Consultant	3	3	2	2	2	2	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Space	Engineering	Consultant	2	2	2	2	2	3	Yes	No	Positive
Not a protected category > 45 years	Male	Scientific Field	High school graduation	Railway	Engineering	Team Leader	3	2	3	2	2	2	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Others	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Defence	Engineering	Consultant	3	3	2	3	3	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Unemployed	Not Specified	Not Specified	3	3	2	1	1	3	Yes	Yes	Positive
Not a protected category 31 - 35 years	Female	Engineering	Five-year degree	Life sciences	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Male	Engineering	Three-year degree	Unemployed	Tech Consulting & Solutions	Advanced Consultant	3	2	2	3	1	3	Yes	Yes	Positive

Not a protected category 26 - 30 years	Female	Engineering	Three-year degree	Unemployed	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Negative
Not a protected category 31 - 35 years	Male	Engineering	Five-year degree	Naval	Not Specified	Not Specified	2	2	2	2	2	3	Yes	Yes	Positive